

Rocky(Oracle Linux) 8 환경에 oracle 19c 설치

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▼ Devices	
Memory	2 GB
Processors	1
Hard Disk (SCSI)	60 GB
CD/DVD (IDE)	Auto detect
Network Adapter	Bridged (Autom...
Display	Auto detect

1. Linux 환경 설정

step 1 oracle 관리 계정 및 그룹 생성

여기서 사용되는 리눅스 설치 계정과 그룹은 오라클 관리 계정 및 그룹으로 사용되고 편의상 홈 디렉토리는 ORACLE_BASE 디렉토리로 사용됨으로 계정의 이름, 디렉토리 등을 설정할 때 주의한다.

호스트명, IP : ora19c, 192.168.10.19

설치 계정 : ora19c (UID : 1900)

소속 그룹 : dba (GID : 1900)

홈 디렉토리 : /home/ora19c

\$ORACLE_BASE : /app/ora19c

\$ORACLE_HOME : /app/ora19c/19c

설치 과정 중에 \$ORACLE_BASE는 /app/ora19c를 사용하는데 이로 인해 oraInventory 디렉토리가 /app에 만들어짐으로 ora19c 계정이 /app 디렉토리에 대해서 소유권과 쓰기 권한을 갖도록 설정해야 한다.

=====

```
# groupadd -g 1900 dba
```

```
# useradd -g dba -u 1900 ora19c
```

```
# passwd ora19c
```

Changing password for user ora19c.

New UNIX password:

Retype new UNIX password:

```
# mkdir -p /app/ora19c/19c
```

```
# mkdir -p /app/oraInventory
```

```
# chown -R ora19c.dba /app/ora19c
```

```
# chown -R ora19c.dba /app/oraInventory
```

```
# chgrp -R dba /app
```

```
# chmod -R 775 /app
```

```
# ls -al /app
```

```
drwxr-xr-x  3 ora19c dba   20  2월  8 23:01 .
```

```
dr-xr-xr-x. 18 root    root 235  2월  8 23:01 ..
drwxr-xr-x  2 ora19c dba    6  2월  8 23:01 ora19c
```

step 2 리눅스 설정

```
=====
# vi /etc/hosts
.....
192.168.10.19    DB19.itclass.co.kr    DB19

# vi /etc/selinux/config    ← selinux 사용을 정지한다.
.....
# SELINUX=enforcing
SELINUX=disabled
.....

# sestatus
SELinux status:                disabled

# systemctl disable --now firewalld.service
# systemctl is-enabled firewalld.service
# systemctl is-active firewalld.service
=====
```

step 3 추가 패키지 설치

오라클 설치에 반드시 필요한 패키지들이다. 본서에서 제공한 리눅스 설치 매뉴얼에 따라 설치된 리눅스의 경우 아래 (설치)라고 표시한 몇몇 패키지를 제외하고 대부분에서는 이미 설치되어있다. 반드시 설치 여부를 확인한다.

```
# dnf -y install ksh libaio-devel glibc-devel libstdc++-devel gcc-c++ libnsd wget
# dnf install -y https://yum.oracle.com/repo/OracleLinux/OL7/latest/x86_64/getPackage/compat-libcap1-1.10-7.el7.x86_64.rpm
# dnf install -y https://yum.oracle.com/repo/OracleLinux/OL7/latest/x86_64/getPackage/compat-libstdc++-33-3.2.3-72.el7.x86_64.rpm

# dnf install -y https://yum.oracle.com/repo/OracleLinux/OL8/appstream/x86_64/getPackage/oracle-database-preinstall-19c-1.0-2.el8.x86_64.rpm
```

또는

```
=====
# vi /etc/yum.repos.d/ol8_appstream.repo
[ol8_appstream]
name=Oracle Linux 8 Application Stream ($basearch)
baseurl=https://yum.oracle.com/repo/OracleLinux/OL8/appstream/x86_64
gpgcheck=0
enabled=1
```

```
# dnf install -y oracle-database-preinstall-19c.x86_64
```

step 4 Ora19c 계정 설정

이전의 설정이나 추가 설치는 모두 root 계정에서 수행했지만 다음의 설정은 ora19c 계정 즉 설치될 오라클의 관리 계정에서 수행한다.

특히 ORACLE_BASE, ORACLE_HOME, ORACLE_SID, TNS_ADMIN 등은 오라클을 운영하는데 매우 중요한 설정이므로 주의 한다.

```
$ vi ~/.bash_profile
```

```
.....
.....
# oracle setup
export TMP=/tmp
export TMPDIR=$TMP
export ORACLE_OWNER=ora19c
export ORACLE_BASE=/app/ora19c
export ORACLE_HOME=/app/ora19c/19c
export ORACLE_SID=DB19
export TNS_ADMIN=$ORACLE_HOME/network/admin
export PATH=$PATH:$ORACLE_HOME/bin:$ORACLE_HOME:/usr/bin:
export NLS_LANG=AMERICAN_AMERICA.AL32UTF8
export ORACLE_HOSTNAME=DB19.itclass.co.kr
export CV_ASSUME_DISTID=RHEL7.6          ← RedHat 계열 리눅스는 아직 7까지만 지원됨
```

ora19c 계정으로 재 로그인 후 설정을 확인한다.

```
$ env | grep ORACLE
```

```
ORACLE_OWNER=ora19c
ORACLE_SID=DB19
ORACLE_BASE=/app/ora19c
ORACLE_HOME=/app/ora19c/19c
```

2. 설치용 패키지 준비

step 1 압축 해제

오라클 원본 파일은 오라클사 홈페이지에서 다운로드 받을 수 있다. 다음 파일을 \$ORACLE_HOME(/app/ora19c/19c) 에 다운받아 압축을 해제한다. 파일명은 LINUX.X64_193000_db_home.zip 이다.

<https://www.oracle.com/database/technologies/oracle19c-linux-downloads.html>

<https://www.oracle.com/database/technologies/oracle19c-windows-downloads.html>
wget http://192.168.10.11/data/down/db/LINUX.X64_193000_db_home.zip

다운받은 파일은 unzip을 이용 압축을 풀어준다.

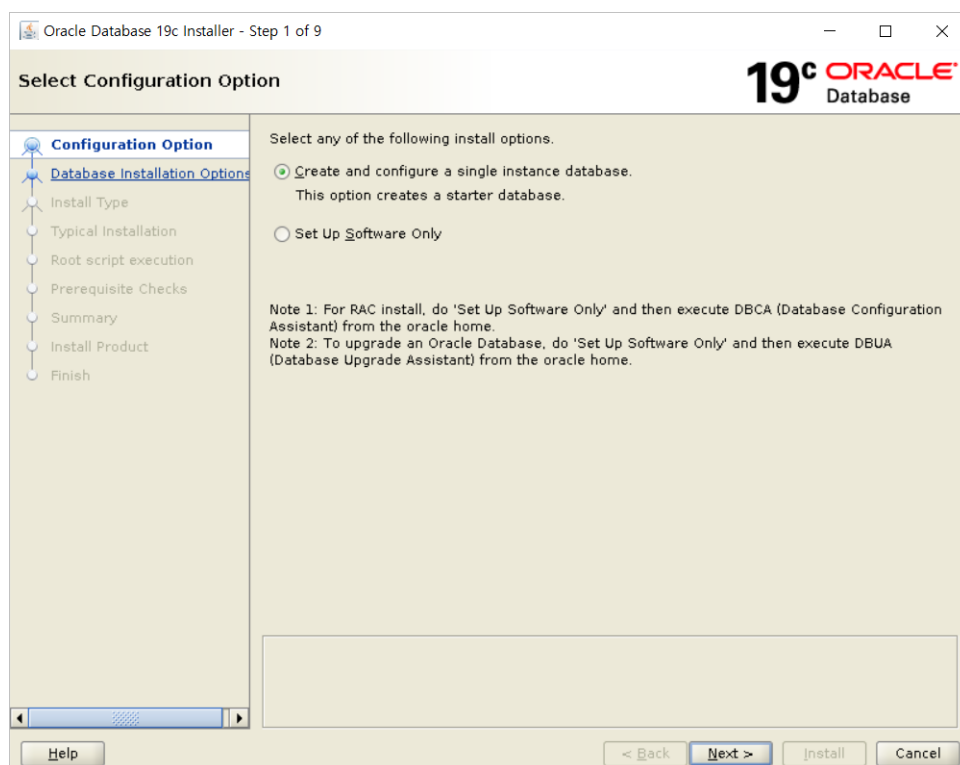
```
[ora19c@ora19c 19c]$ unzip LINUX.X64_193000_db_home.zip
```

step 2 인스톨러 실행

Redhat 계열 linux8을 지원하지 않음으로 언어 설정을 취소하고 인스톨러를 실행한다.

```
[ora19c@ora19c 19c]$ unset LANG  
[ora19c@ora19c 19c]$ ./runInstaller
```

3. Universal Installer



Oracle Database 19c Installer - Step 2 of 8

Select System Class

19c ORACLE Database

Configuration Option

System Class

Typical Installation

Root script execution

Prerequisite Checks

Summary

Install Product

Finish

☒ Desktop class
Choose this option if you are installing on a laptop or desktop class system. This option includes a starter database and allows minimal configuration.

☐ Server class
Choose this option if you are installing on a server class system, which Oracle defines as a system used in a production data center. This option allows for more advanced configuration options.

Help < Back Next > Install Cancel

Oracle Database 19c Installer - Step 3 of 8

Typical Install Configuration

19c ORACLE Database

Configuration Option

System Class

Typical Installation

Root script execution

Prerequisite Checks

Summary

Install Product

Finish

Perform full database installation with basic configuration.

Oracle base: /app/ora19c Browse...

Software location: /app/ora19c/19c

Database file location: /app/ora19c/oradata Browse...

Database edition: Enterprise Edition

Character set: Unicode (AL32UTF8)

OSDBA group: dba

Global database name: DB19

Password: Password Confirm password:

☐ Create as Container database

Pluggable database name: orclpdb

Messages:

Warning: [INS-30011] The ADMIN password entered does not conform to the Oracle recommended standards.

Help < Back Next > Install Cancel

Oracle Database 19c Installer

Warning: [INS-30011] The ADMIN password entered does not conform to the Oracle recommended standards.

Are you sure you want to continue ?

Yes No Details

Oracle Database 19c Installer - Step 4 of 9

Create Inventory

19c ORACLE Database

Configuration Option
System Class
Typical Installation
Create Inventory
Root script execution
Prerequisite Checks
Summary
Install Product
Finish

You are starting your first installation on this host. Specify a directory for installation metadata files (for example, install log files). This directory is called the "inventory directory". The installer automatically sets up subdirectories for each product to contain inventory data. The subdirectory for each product typically requires 150 kilobytes of disk space.

Inventory Directory:

Specify an operating system group whose members have write permission to the inventory directory (orainventory).

orainventory Group Name:

Oracle Database 19c Installer - Step 5 of 9

Root script execution configuration

19c ORACLE Database

Configuration Option
System Class
Typical Installation
Create Inventory
Root script execution
Prerequisite Checks
Summary
Install Product
Finish

During the software configuration, certain operations have to be performed as "root" user. You can choose to have the installer perform these operations automatically by specifying inputs for one of the options below. The input specified will also be used by the installer to perform additional prerequisite checks.

☒ Automatically run configuration scripts

☒ Use "root" user credential

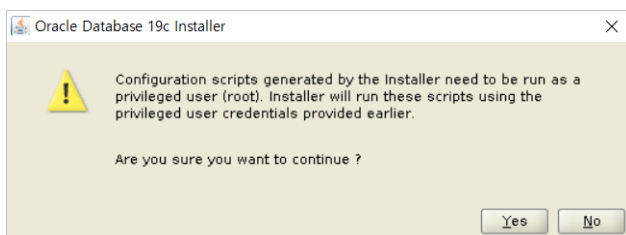
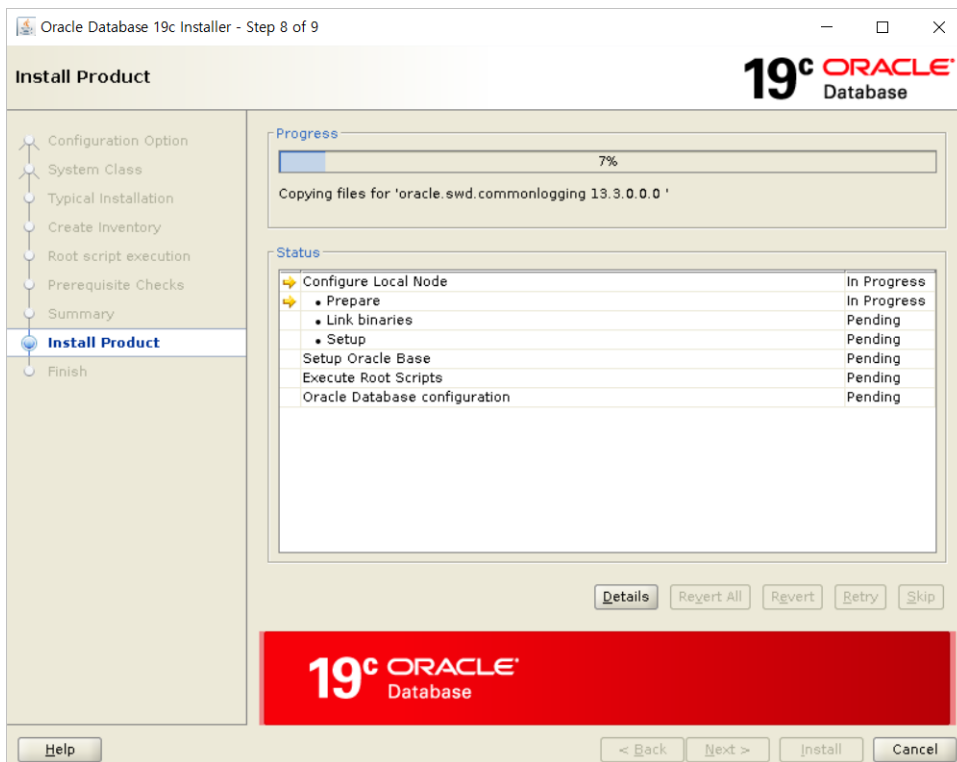
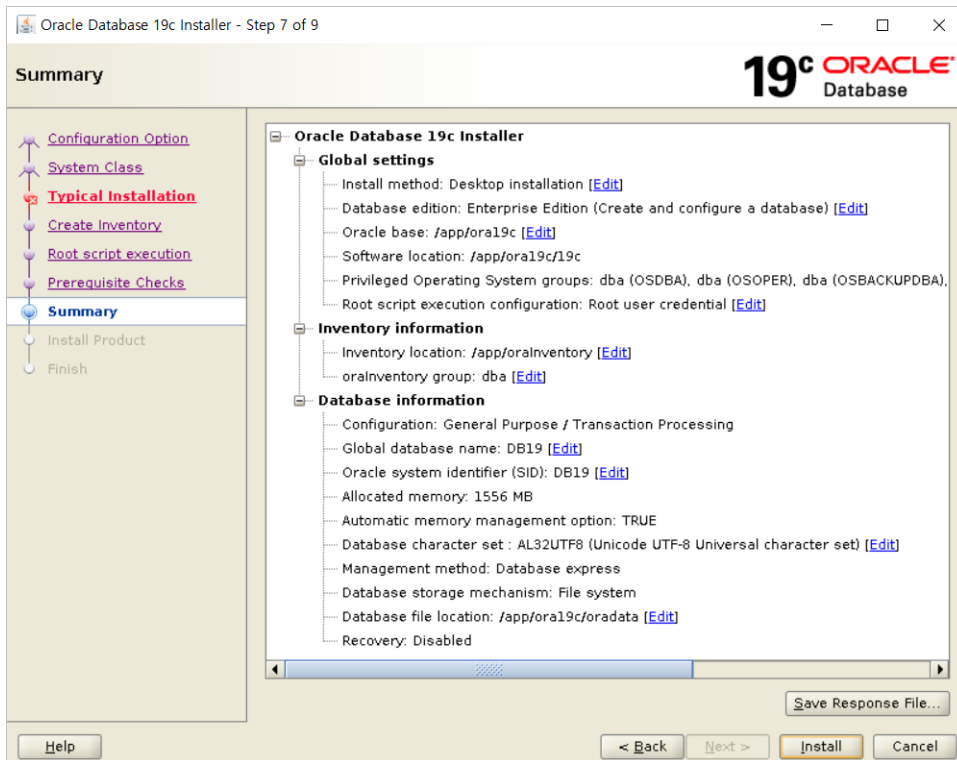
Password :

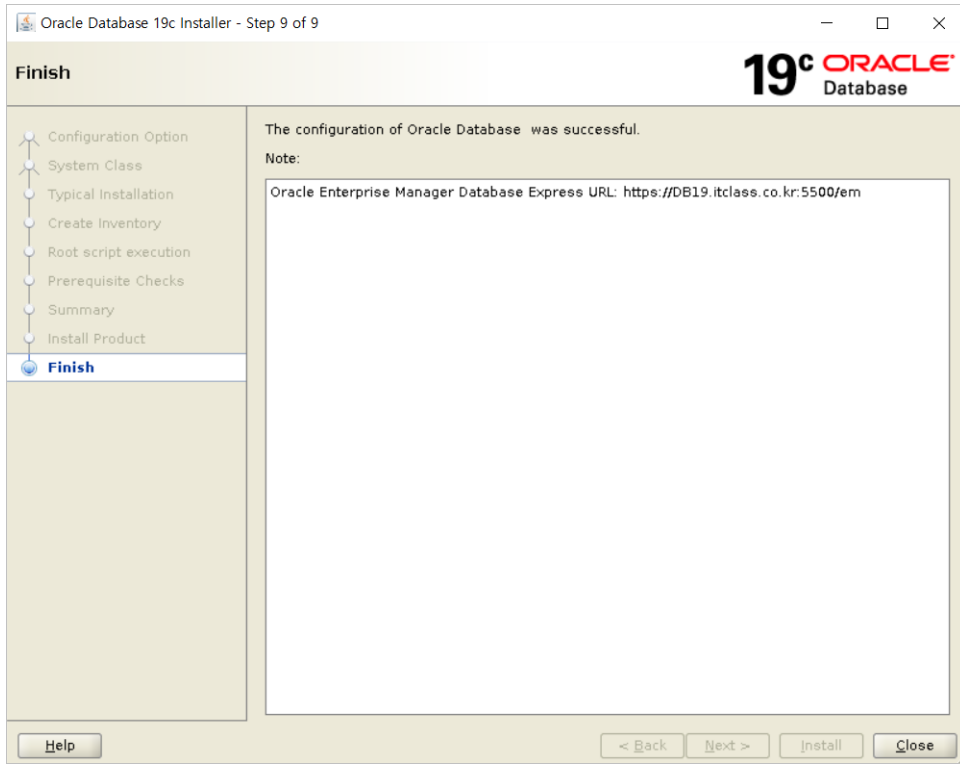
☐ Use sudo

Program path :

User name :

Password :





4. 설치 확인

```
=====
$ sqlplus / as sysdba
```

```
SQL*Plus: Release 19.0.0.0.0 - Production on Sat Jun 7 13:23:44 2025
Version 19.3.0.0.0
```

```
Copyright (c) 1982, 2019, Oracle. All rights reserved.
```

```
Connected to:
```

```
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0
```

```
SQL> SELECT status FROM v$instance;
```

```
STATUS
```

```
OPEN
=====
```

5. listener 파일 확인

```
=====
$ cat /app/ora19c/19c/network/admin/listener.ora
# listener.ora Network Configuration File: /app/ora19c/19c/network/admin/listener.ora
ITCLASS
```



```
# Generated by Oracle configuration tools.
```

```
LISTENER =
```

```
  (DESCRIPTION_LIST =
```

```
    (DESCRIPTION =
```

```
      (ADDRESS = (PROTOCOL = TCP)(HOST = DB19.itclass.co.kr)(PORT = 1521))
```

```
      (ADDRESS = (PROTOCOL = IPC)(KEY = EXTPROC1521))
```

```
    )
```

```
  )
```

```
SID_LIST_LISTENER =
```

```
  (SID_LIST =
```

```
    (SID_DESC =
```

```
      (SID_NAME = DB19)
```

```
      (ORACLE_HOME = /app/ora19c/19c)
```

```
    )
```

```
  )
```

```
$ cat /app/ora19c/19c/network/admin/sqlnet.ora
```

```
NAMES.DIRECTORY_PATH= (TNSNAMES, EZCONNECT)
```

```
$ cat <<EOF >> /app/ora19c/19c/network/admin/tnsnames.ora
```

```
oracle =
```

```
  (DESCRIPTION =
```

```
    (ADDRESS = (PROTOCOL = tcp) (HOST = 192.168.##.##) (port = 1521))
```

```
    (CONNECT_DATA = (SID = DB19))
```

```
  )
```

```
EOF
```

```
$ lsnrctl status
```

```
$ lsnrctl start
```

```
=====
```